INTERNSHIP REPORT - TASK 2

Title: Data cleaning and Exploratory Data Analysis(EDA)

🧑‍🎓 Student Details

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- **Task:** 2

🎯 Objective

The objective of this task is to perform **data cleaning and exploratory data analysis (EDA)** on the Titanic dataset in order to identify patterns and factors that influenced passenger survival.

📁 Dataset Description

The Titanic dataset contains passenger information such as:

- Age
- Gender (Sex)
- Passenger Class (Pclass)
- Fare
- Embarkation point
- Survival status (0 = Did not survive, 1 = Survived)

This dataset is widely used for understanding classification problems and data exploration techniques.

🧹 Data Cleaning Process

The following steps were performed to clean the dataset:

- Missing values in **Age** were filled using the mean value
- Missing values in **Embarked** were filled using the mode

- The **Deck column** was removed due to a large number of missing values
- Duplicate records were checked and handled appropriately

📊 Exploratory Data Analysis (EDA)

The following analyses were performed:

- Survival count distribution
- Gender vs Survival comparison
- Passenger Class vs Survival analysis
- Age distribution of passengers
- Fare distribution analysis
- Correlation heatmap between numerical features
- Boxplots and pie charts for visual insights

🔍 Key Insights

From the analysis, the following patterns were observed:

- Most passengers did **not survive** the disaster
- **Females had a significantly higher survival rate** compared to males
- Passengers in **First Class had better survival chances**
- Higher fare values were associated with higher survival probability
- Passenger class had a strong influence on survival outcome

📌 Conclusion

The exploratory data analysis clearly shows that survival on the Titanic was strongly influenced by **gender, passenger class, and fare amount**. These features played a major role in determining the likelihood of survival.